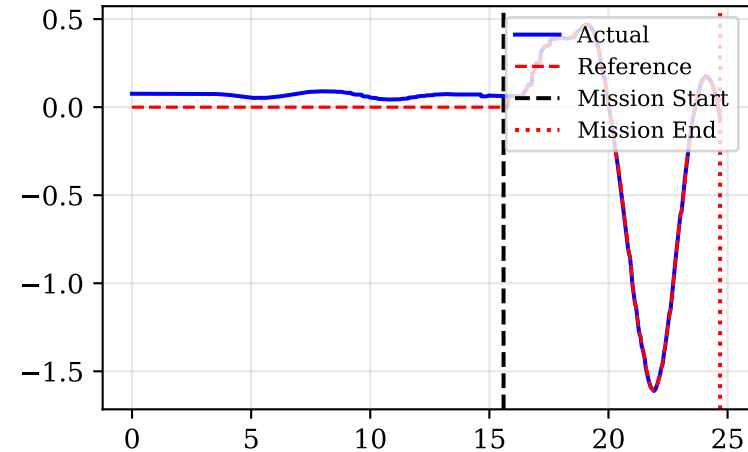
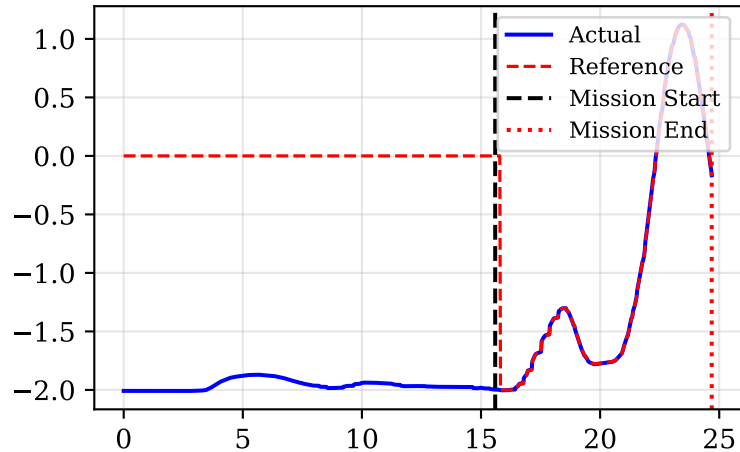


# Drone Position vs MPC optimal trajectory

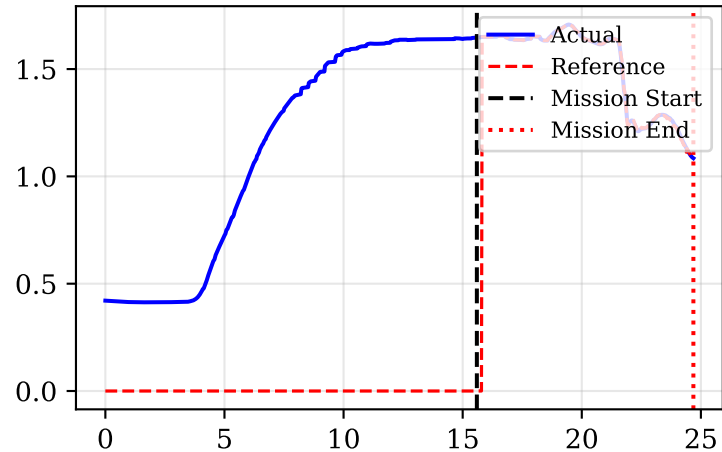
Position X [m]



Position Y [m]

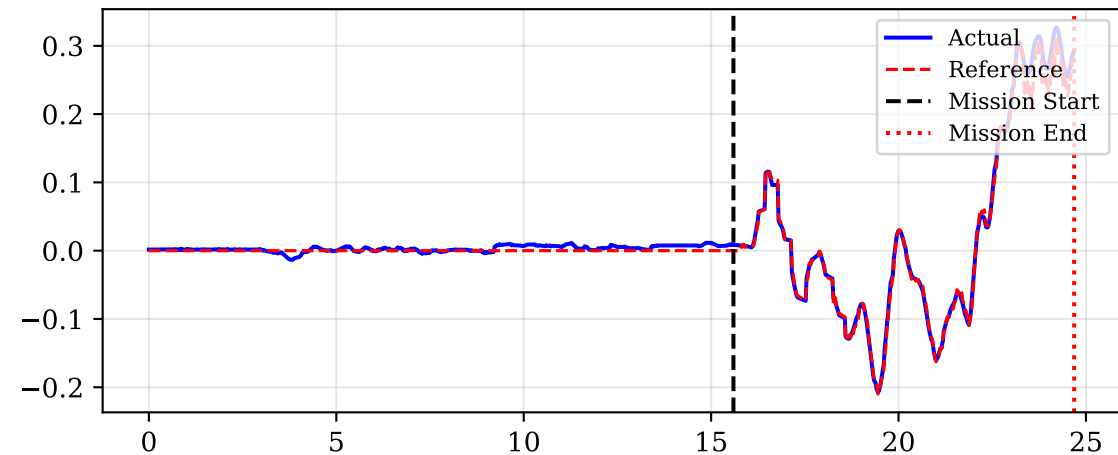


Position Z [m]

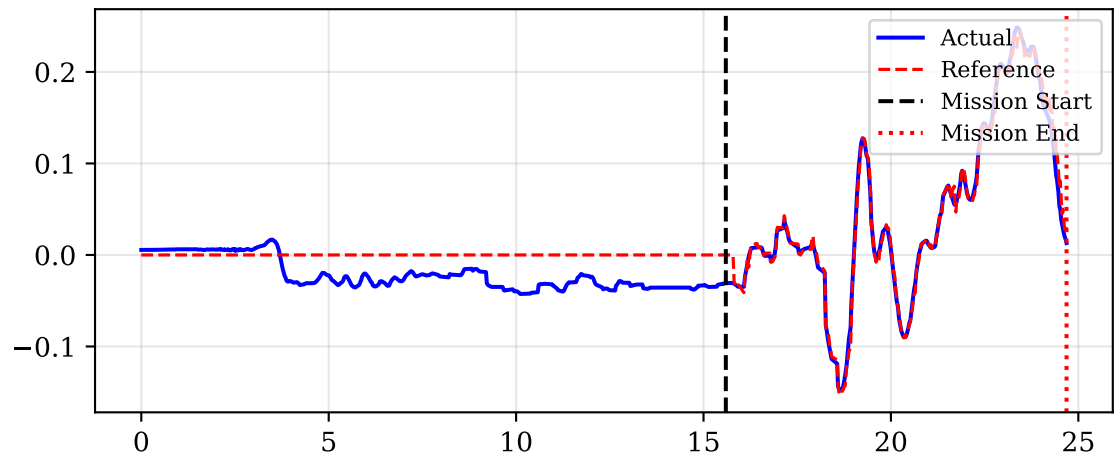


# Drone Orientation (Roll/Pitch)

Roll [rad]

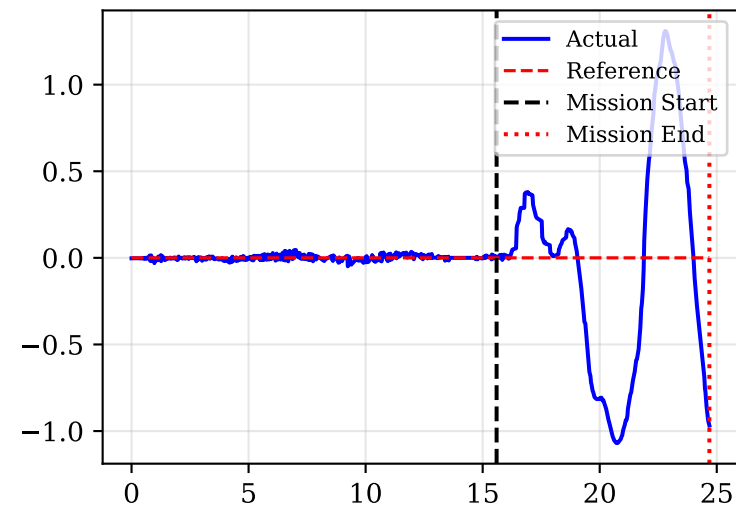


Pitch [rad]

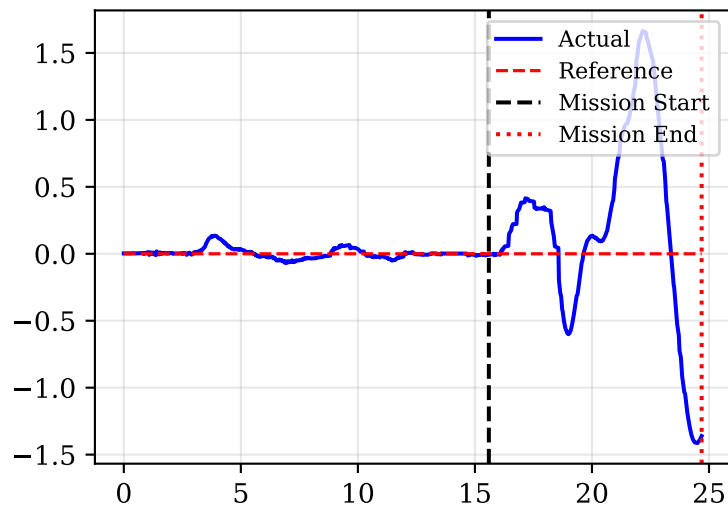


# Drone Velocities vs MPC Reference

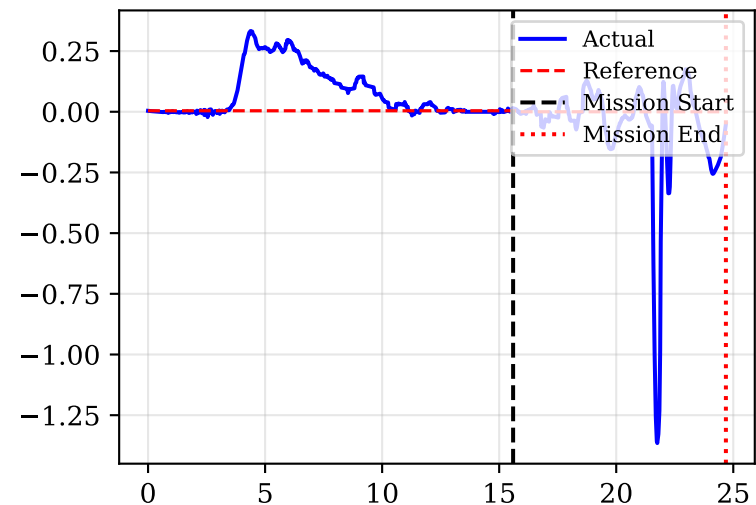
Vel X [m/s]



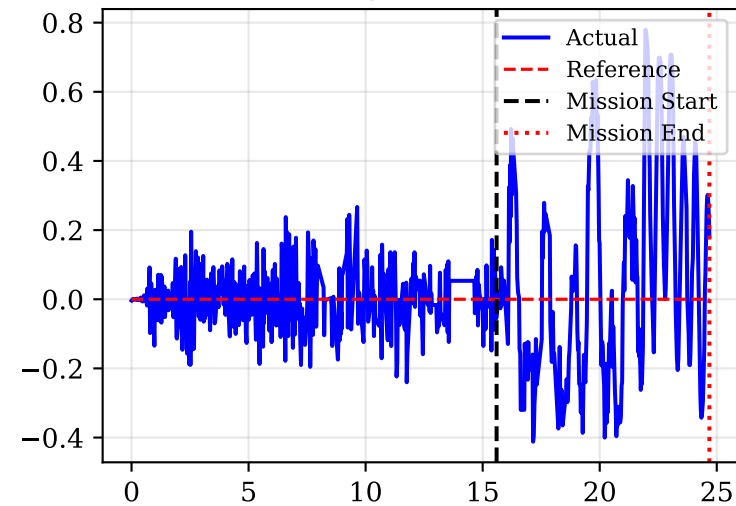
Vel Y [m/s]



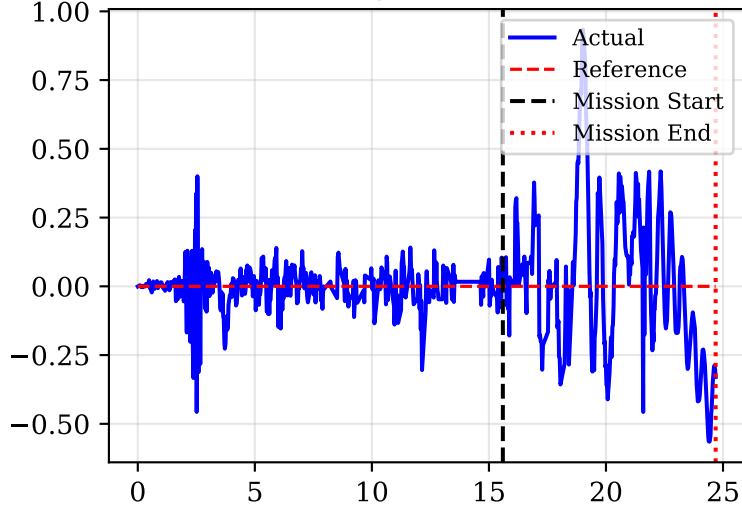
Vel Z [m/s]



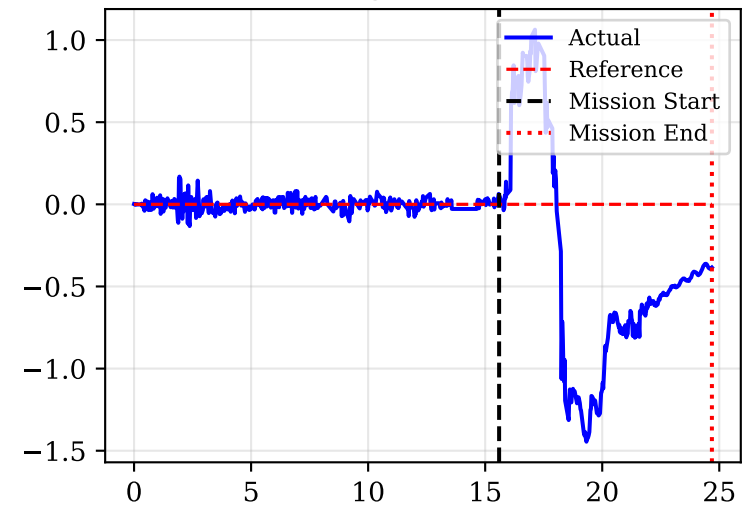
Omega X [rad/s]



Omega Y [rad/s]

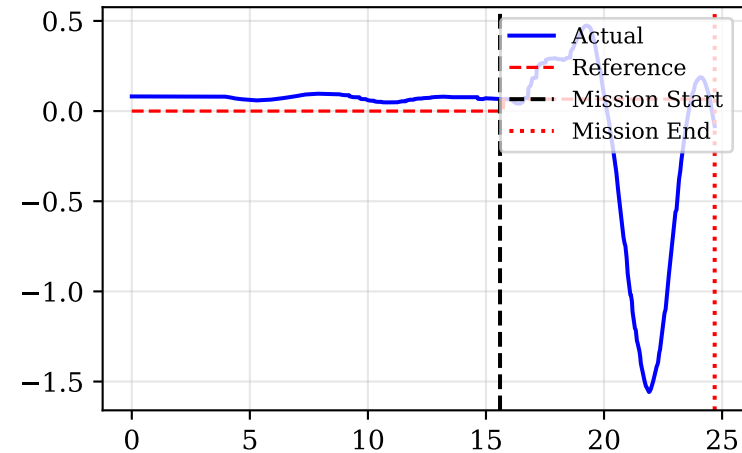


Omega Z [rad/s]

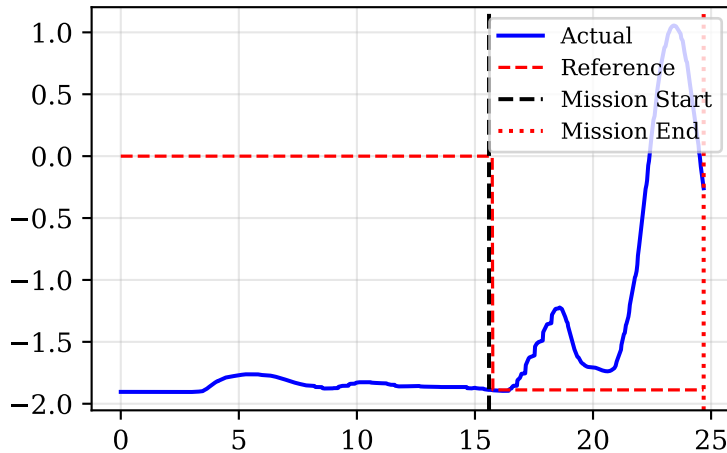


# Camera Cartesian Position vs Derived Target

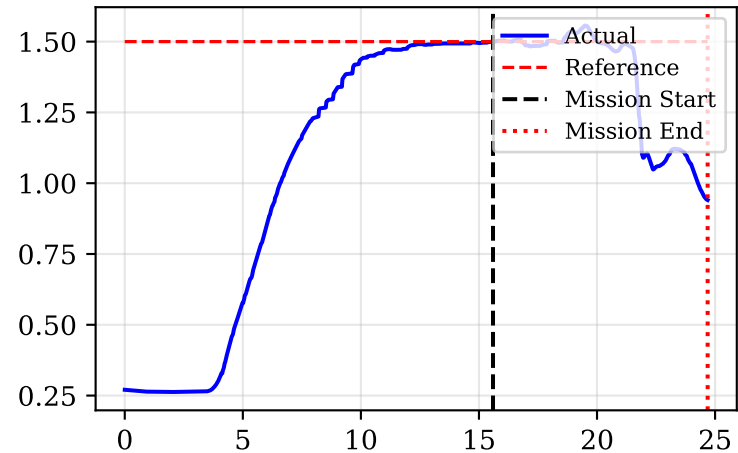
Cam X [m]



Cam Y [m]

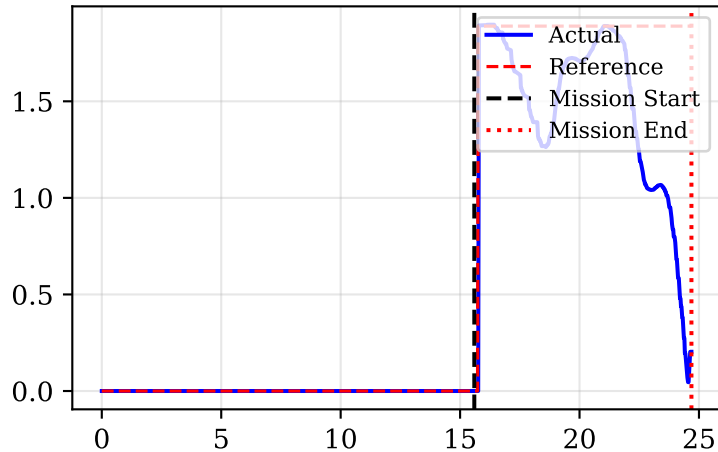


Cam Z [m]

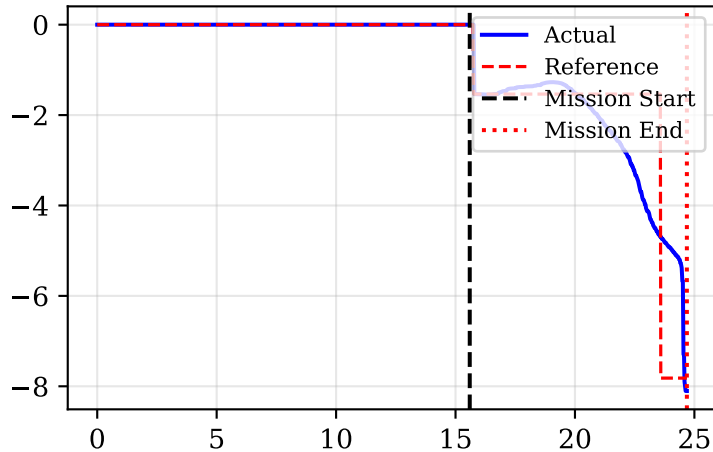


# Cylindrical PoV Tracking (World Frame)

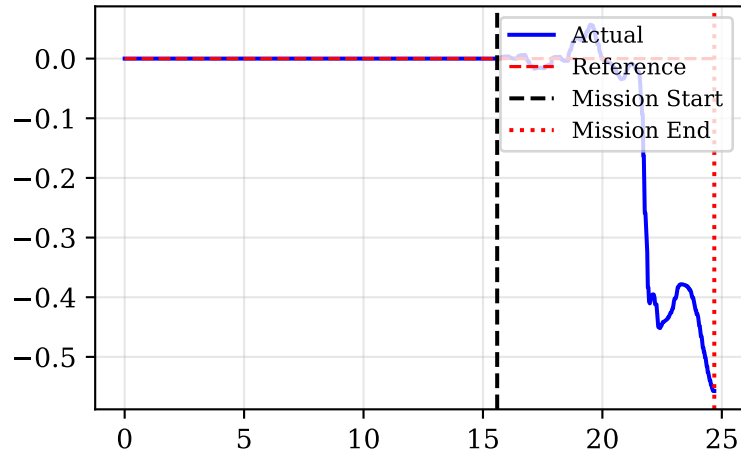
Distance  $r_{\text{cyl}}$  [m]



Azimuth  $\beta$  [rad]

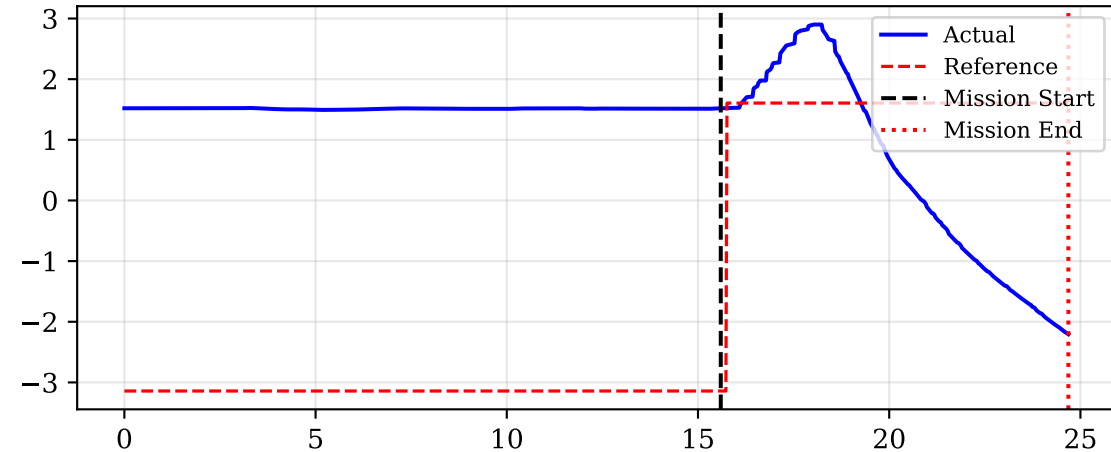


Elevation  $z$  [m]

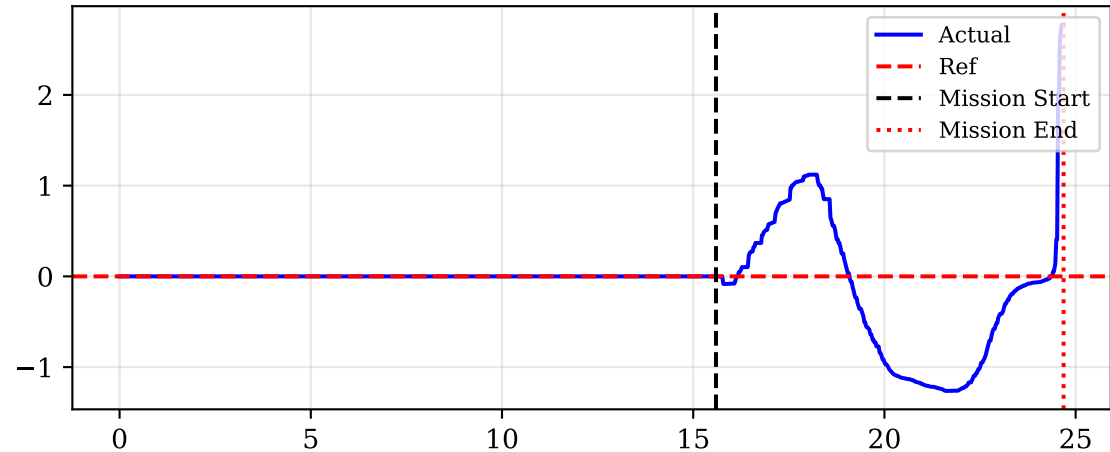


# Yaw Tracking: Drone Pointing Toward Target

## Yaw Actual vs Desired [rad]

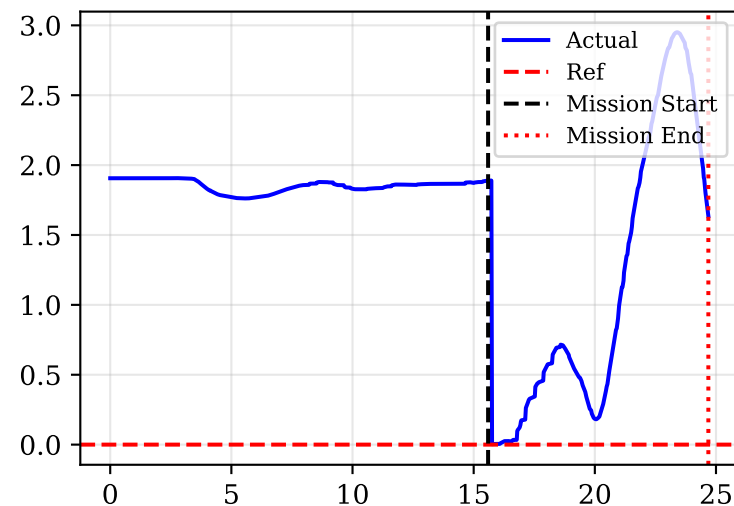


## Yaw Error [rad]

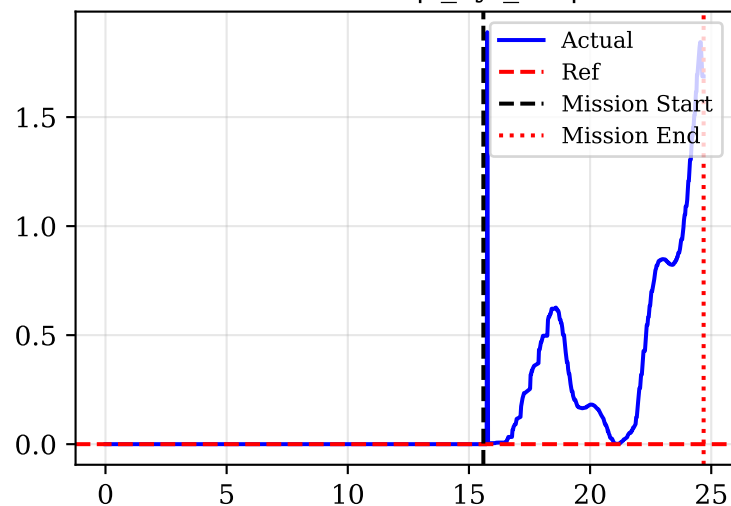


# Primary Tracking Errors (Cylindrical)

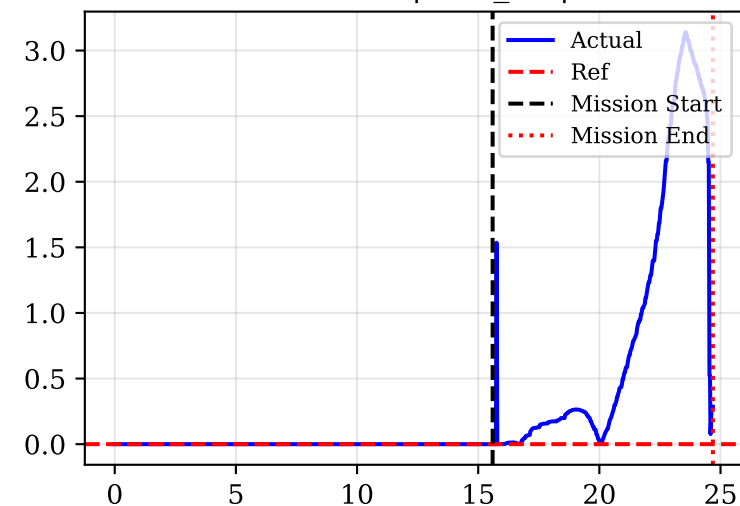
## Cam XY Error [m]



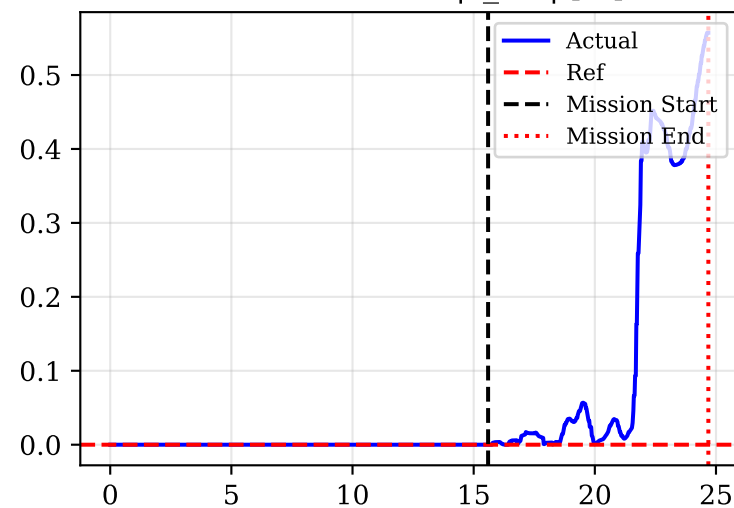
## Distance Error [r\_cyl\_err] [m]



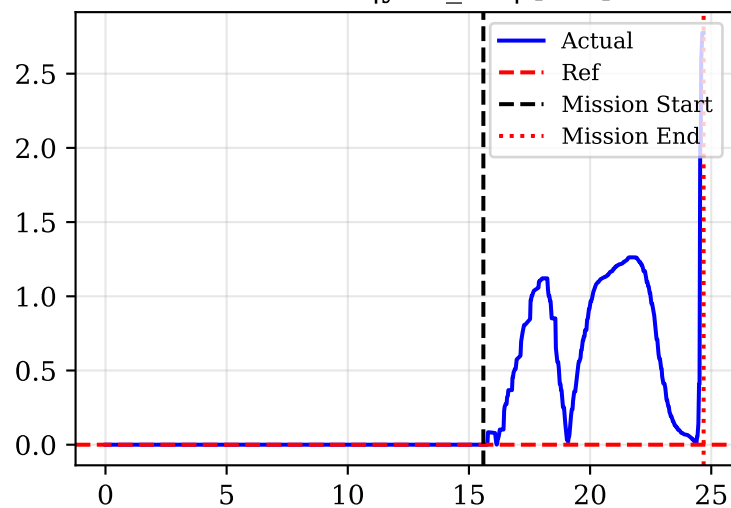
## Azimuth Error [beta\_err] [rad]



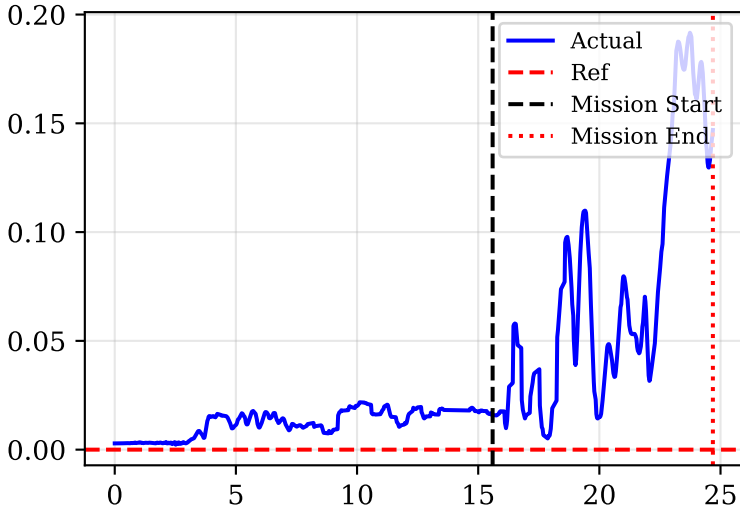
## Elevation Error [z\_err] [m]



## Yaw Error [yaw\_err] [rad]

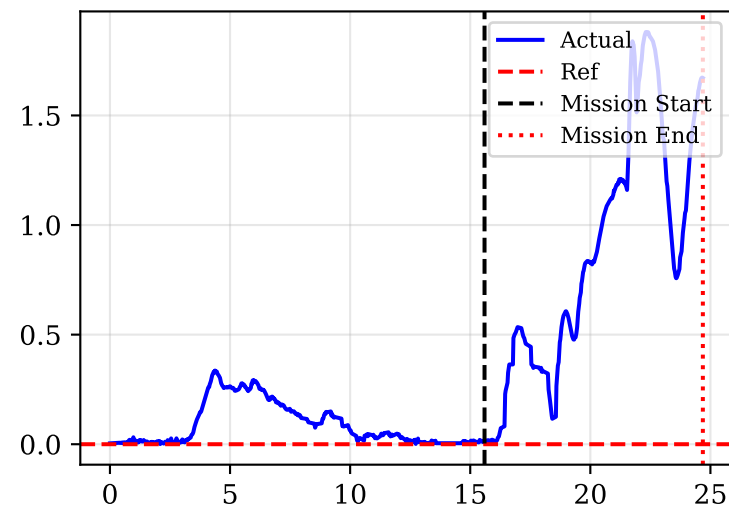


## Norm Roll/Pitch Error

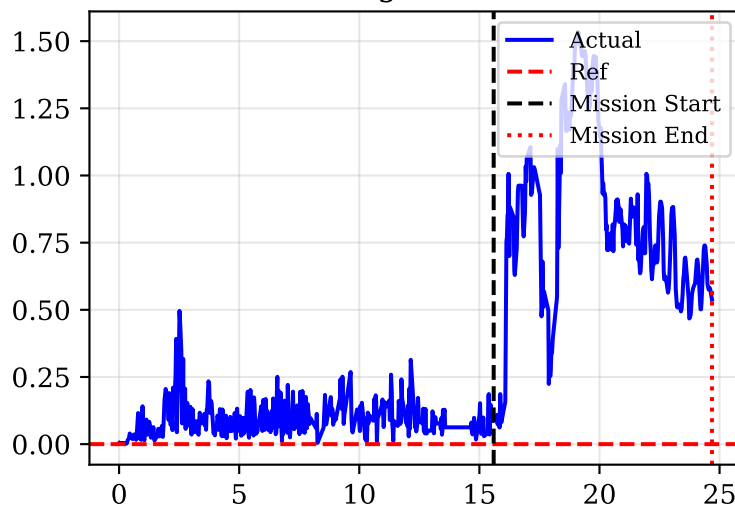


# Dynamic States Errors and Feedforward Derivatives

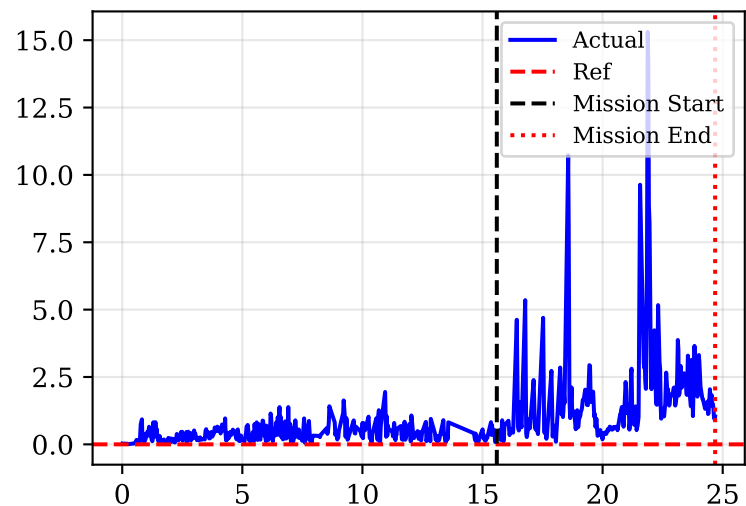
Norm Vel Error [m/s]



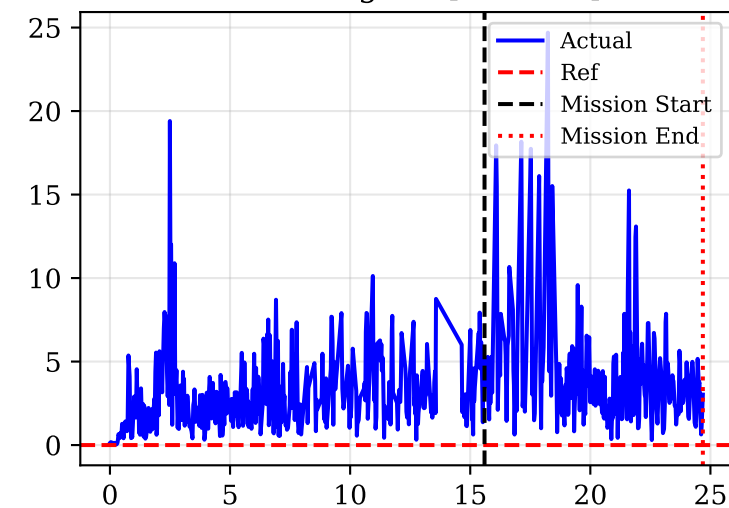
Norm Omega Error [rad/s]



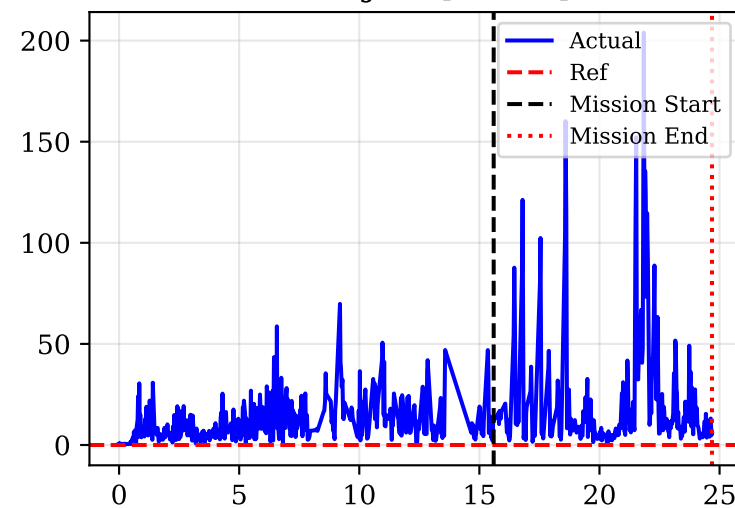
Norm Acc [m/s^2]



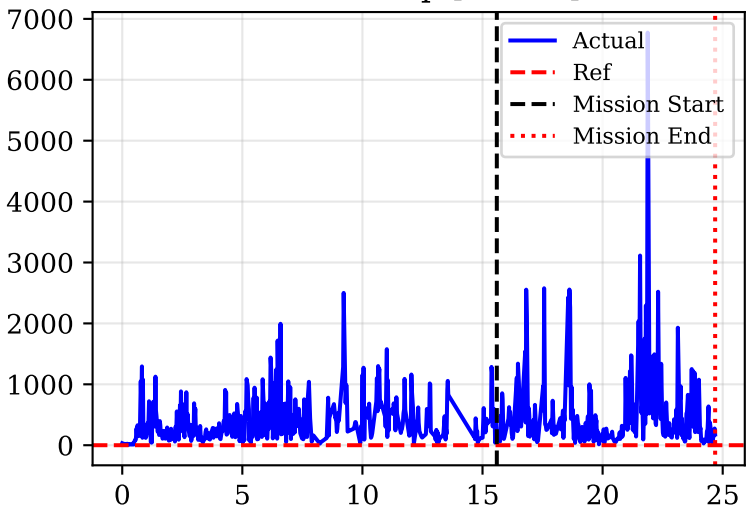
Norm AngAcc [rad/s^2]



Norm Jerk [m/s^3]



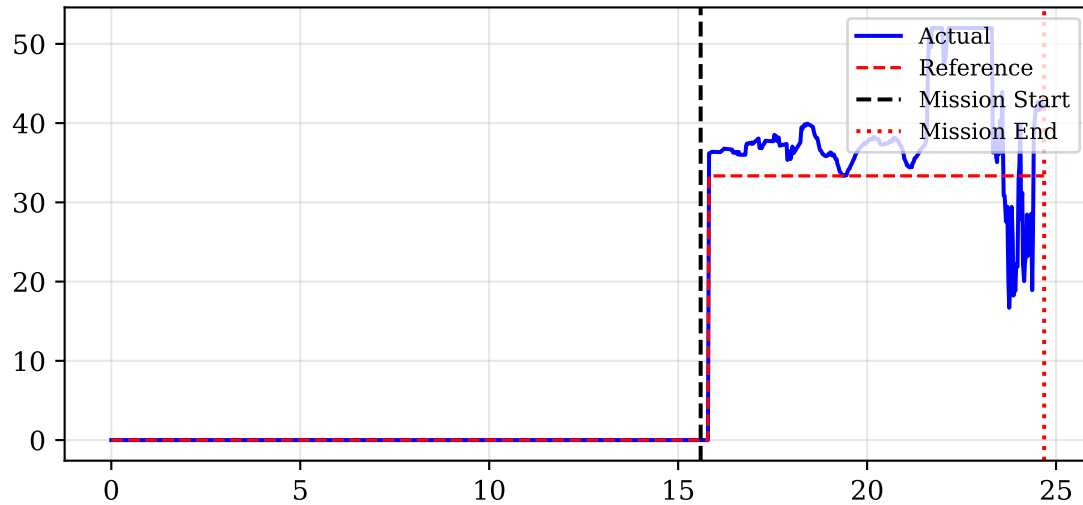
Norm Snap [m/s^4]



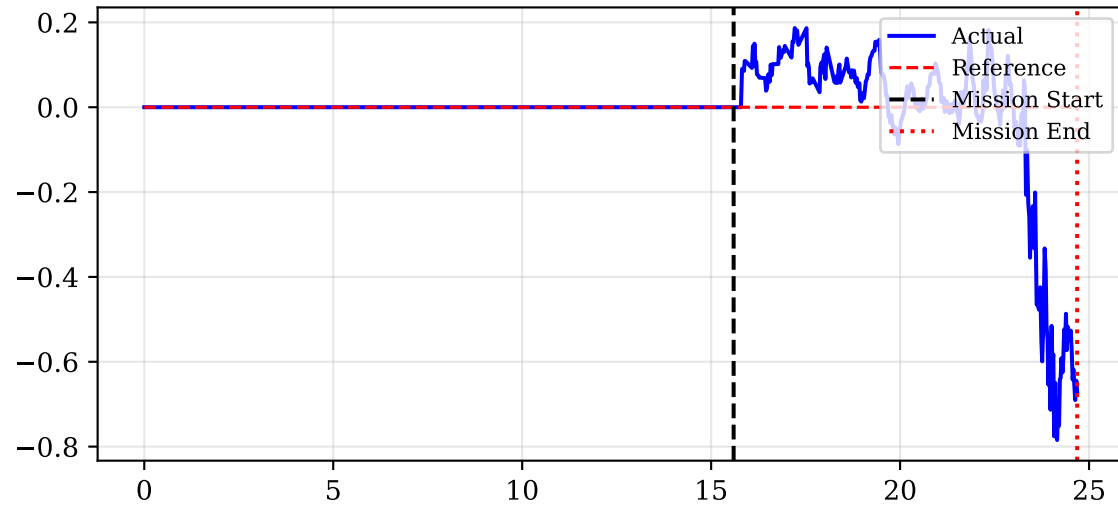


# Control Wrench (Hover Force = 20.24N)

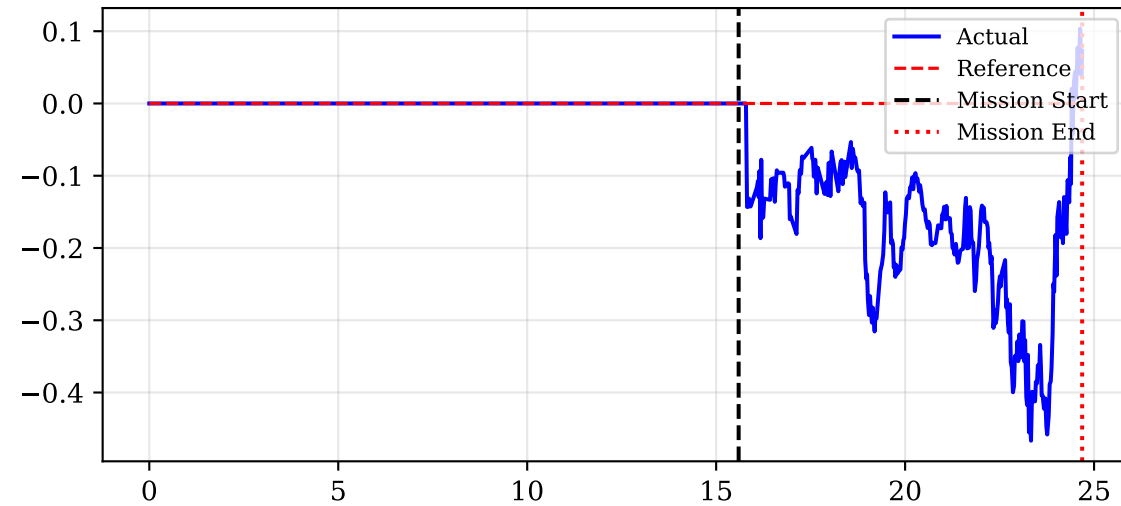
Force Z (Thrust) [N]



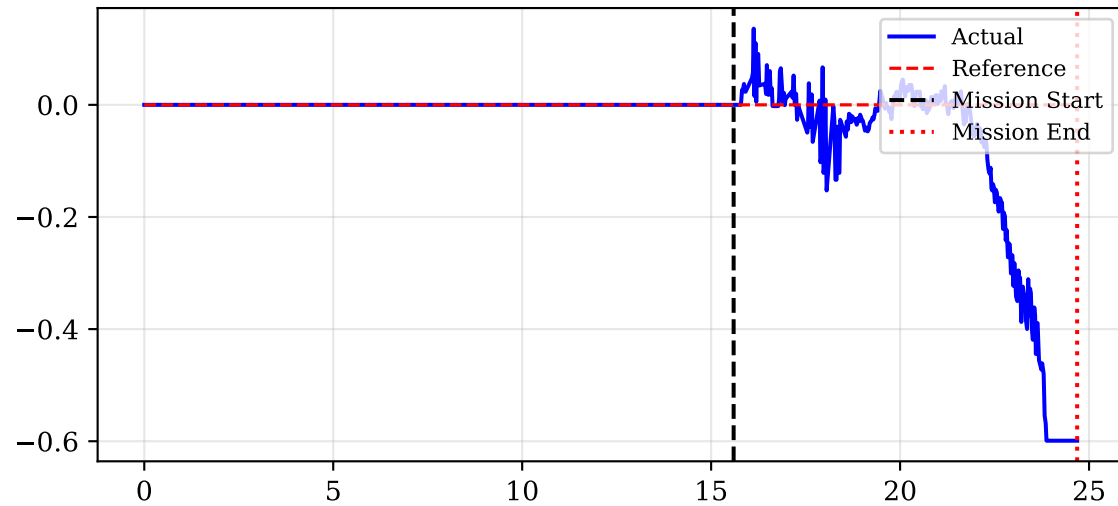
Torque X [Nm]



Torque Y [Nm]

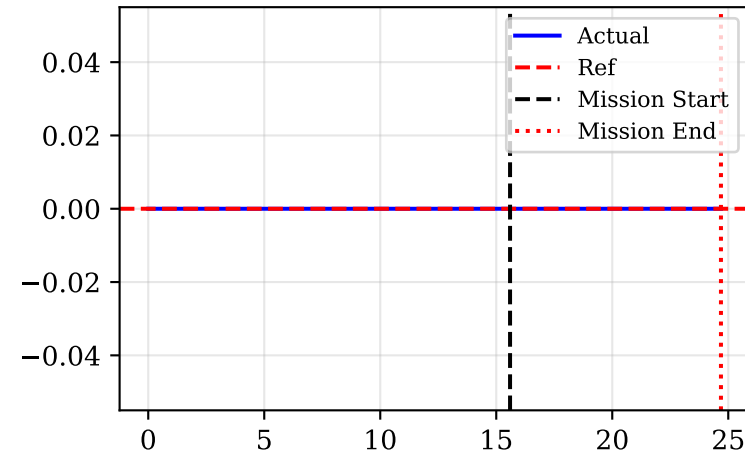


Torque Z [Nm]

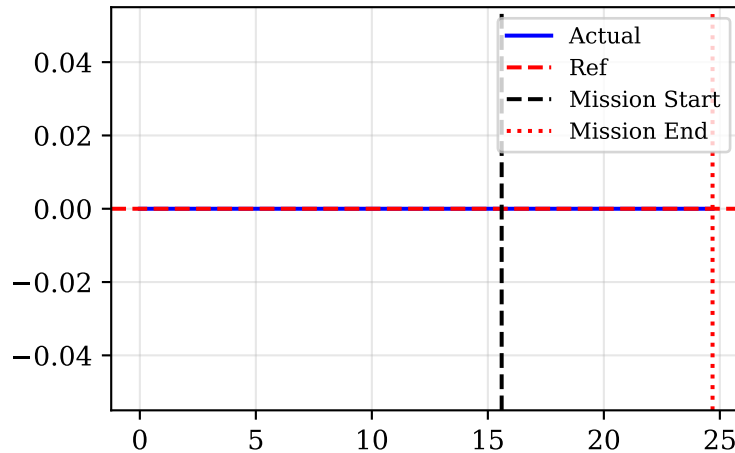


# Haptic Feedback Forces Transmitted to haptic device

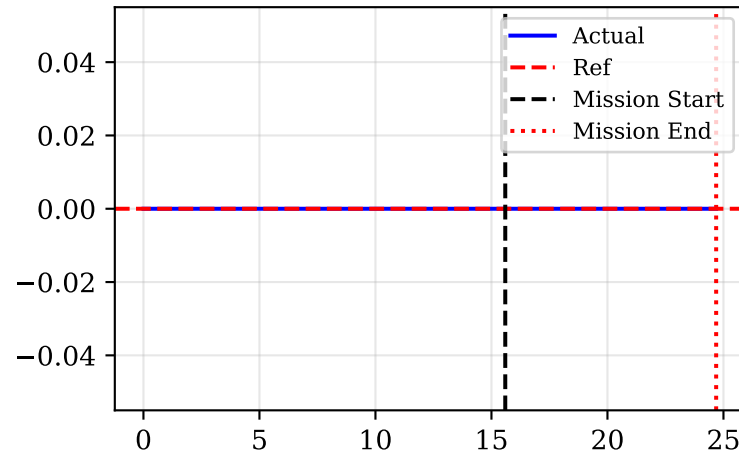
Force X (Zoom) [N]



Force Y (Pan) [N]

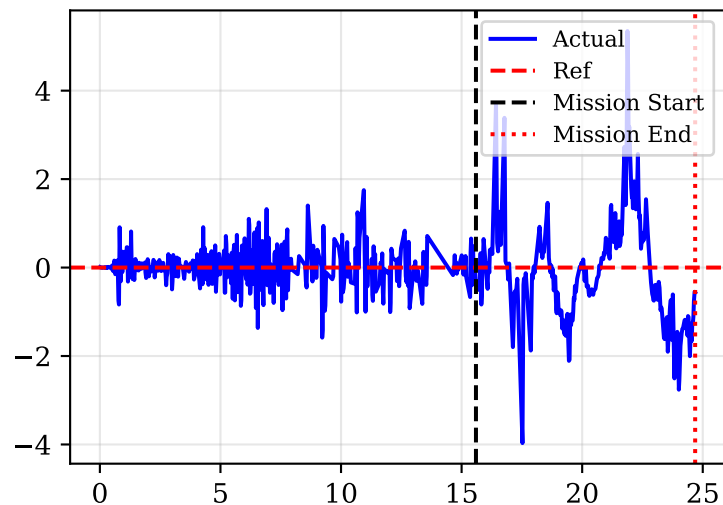


Force Z (Altitude) [N]

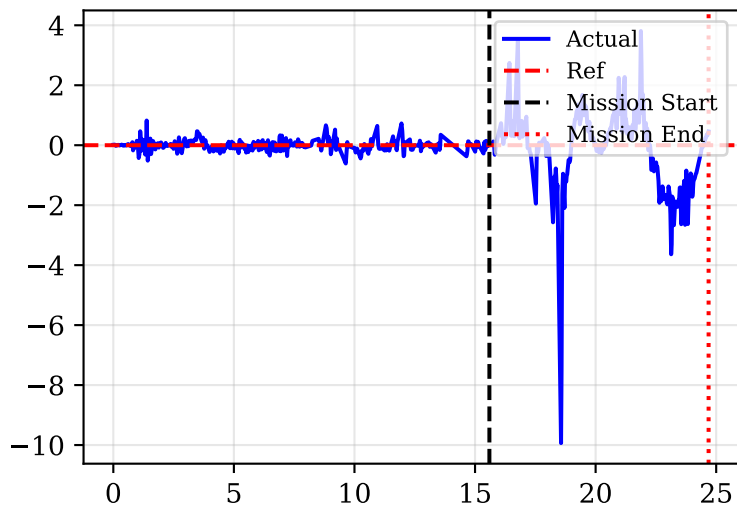


# Drone Linear and Angular Accelerations

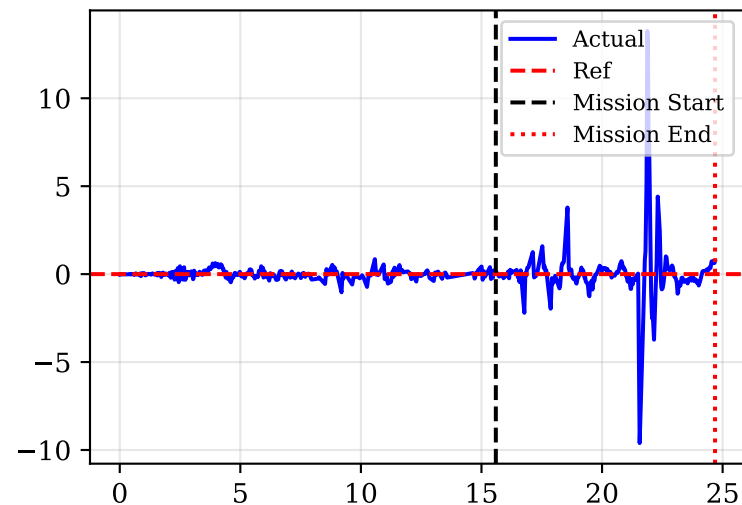
Linear Acc X [m/s<sup>2</sup>]



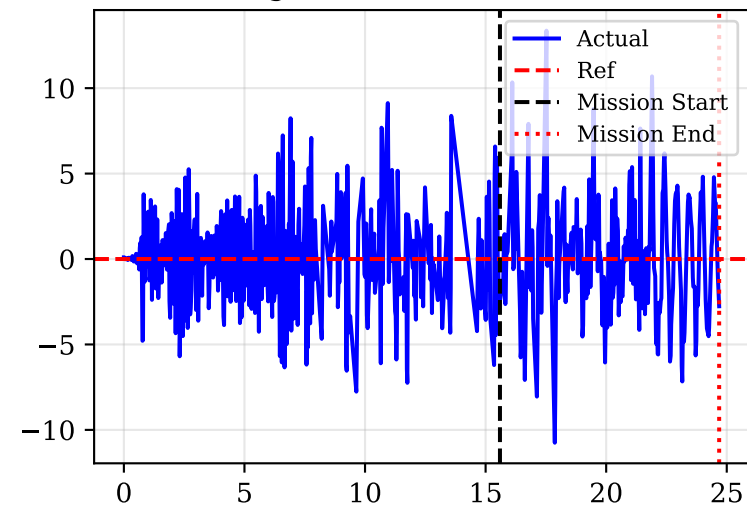
Linear Acc Y [m/s<sup>2</sup>]



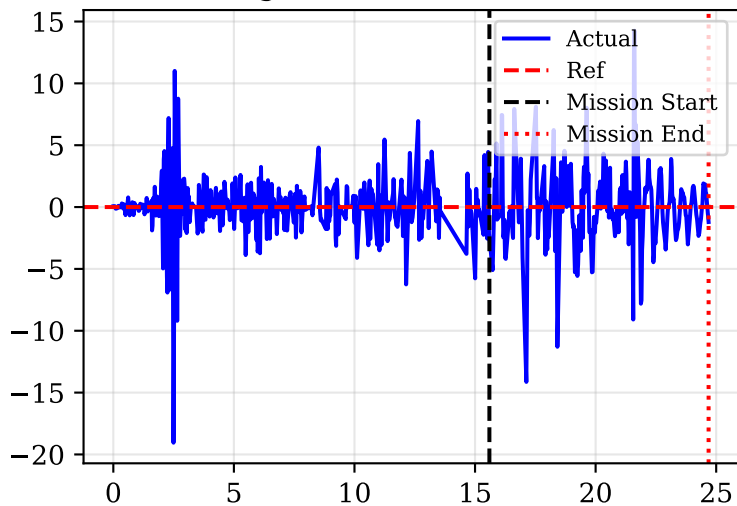
Linear Acc Z [m/s<sup>2</sup>]



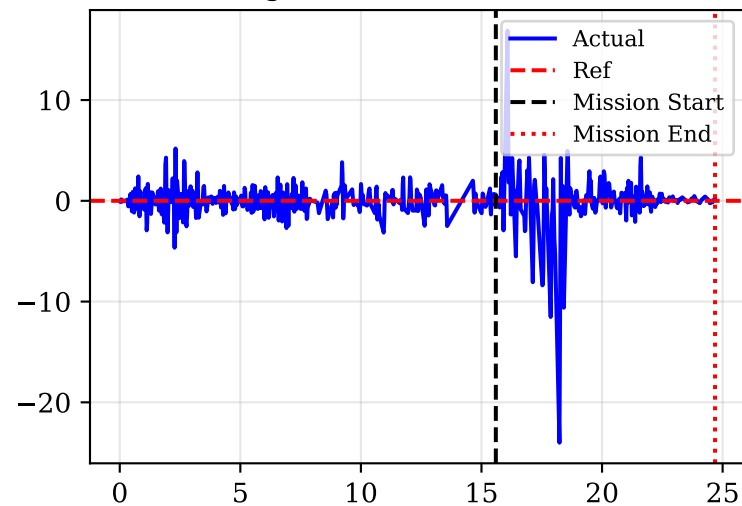
Angular Acc X [rad/s<sup>2</sup>]



Angular Acc Y [rad/s<sup>2</sup>]

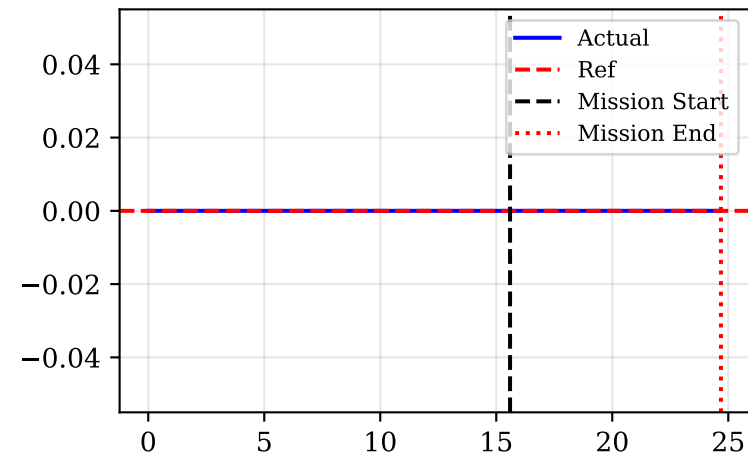


Angular Acc Z [rad/s<sup>2</sup>]

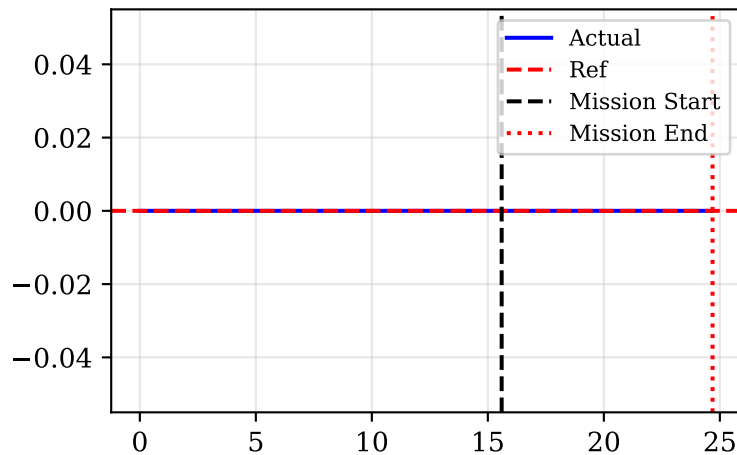


# Peg External Contact Forces (FT Sensor)

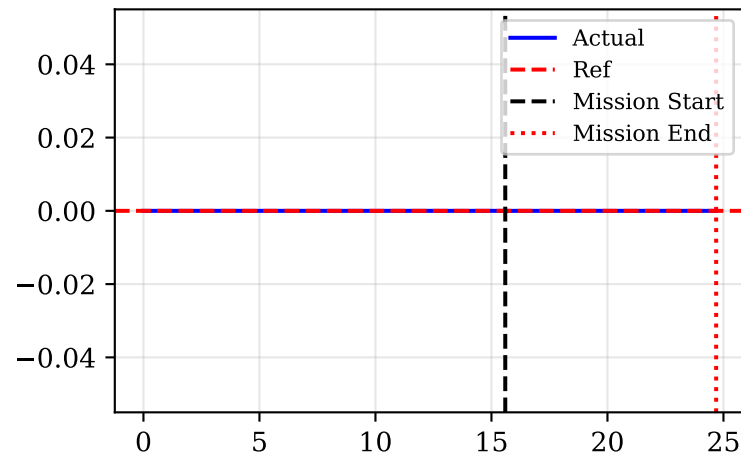
## Force X (Sensor) [N]



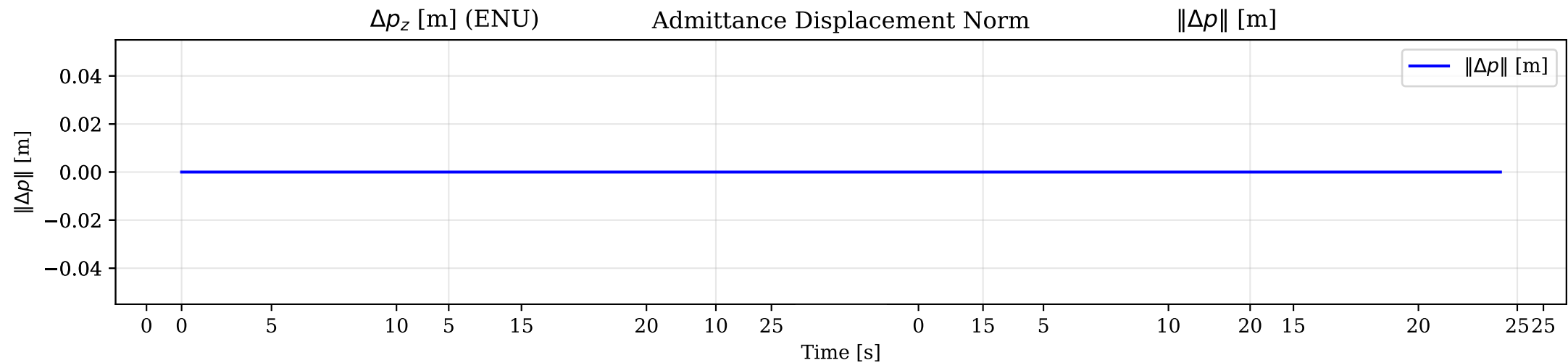
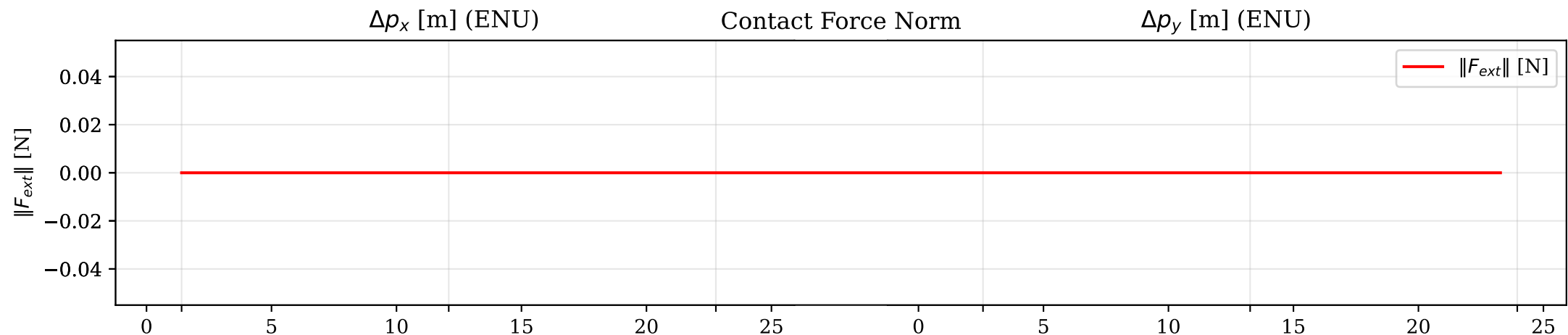
## Force Y (Sensor) [N]



## Force Z (Sensor) [N]

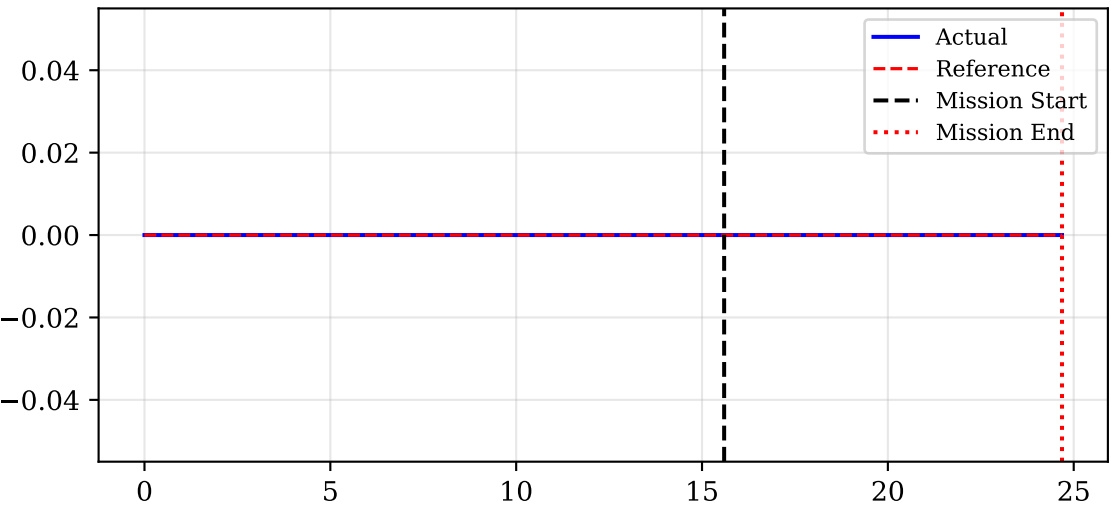


# Admittance Effect: Contact Force vs Position Deviation

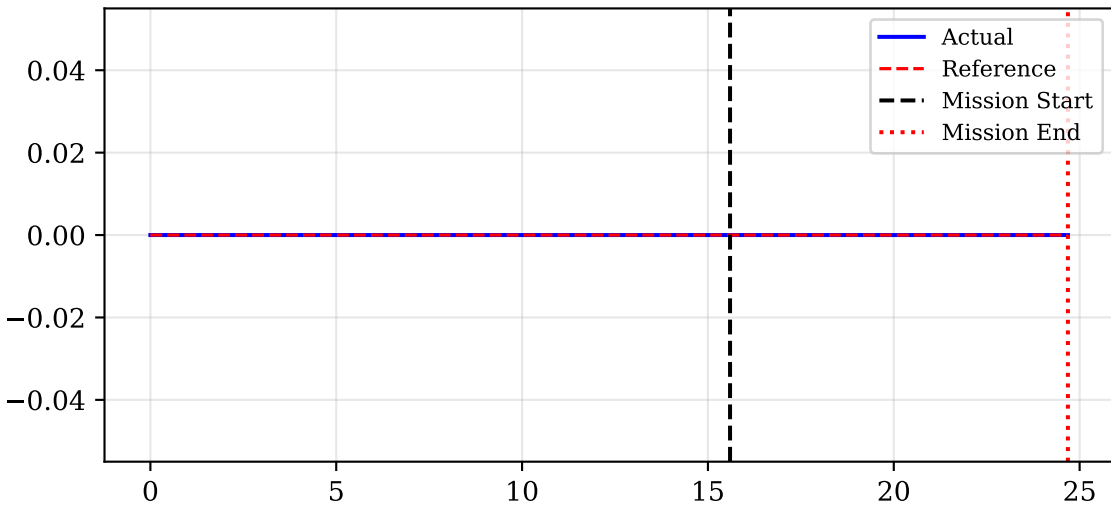


# Interaction Drone Position ENU (Actual vs Planner Reference)

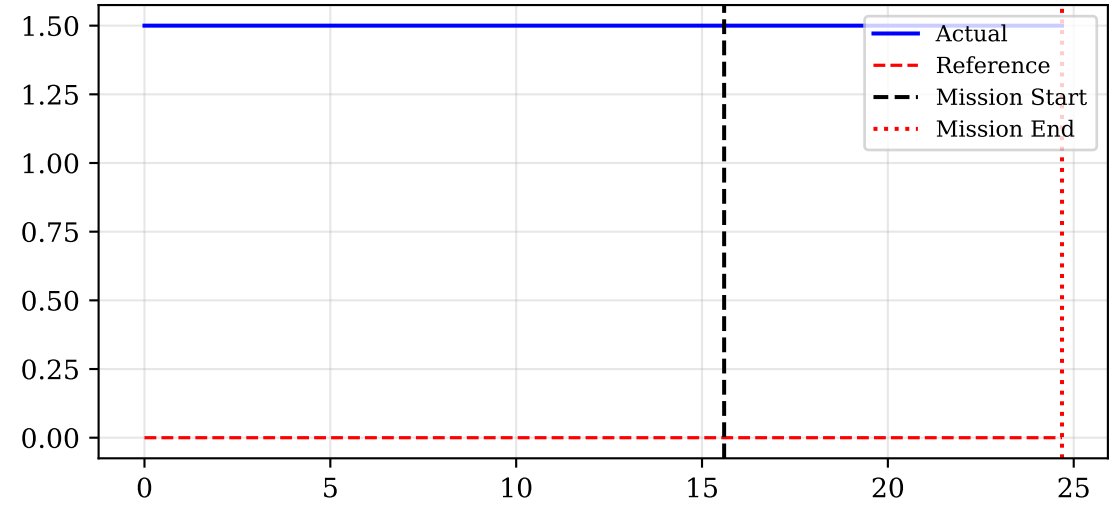
Peg X [m]



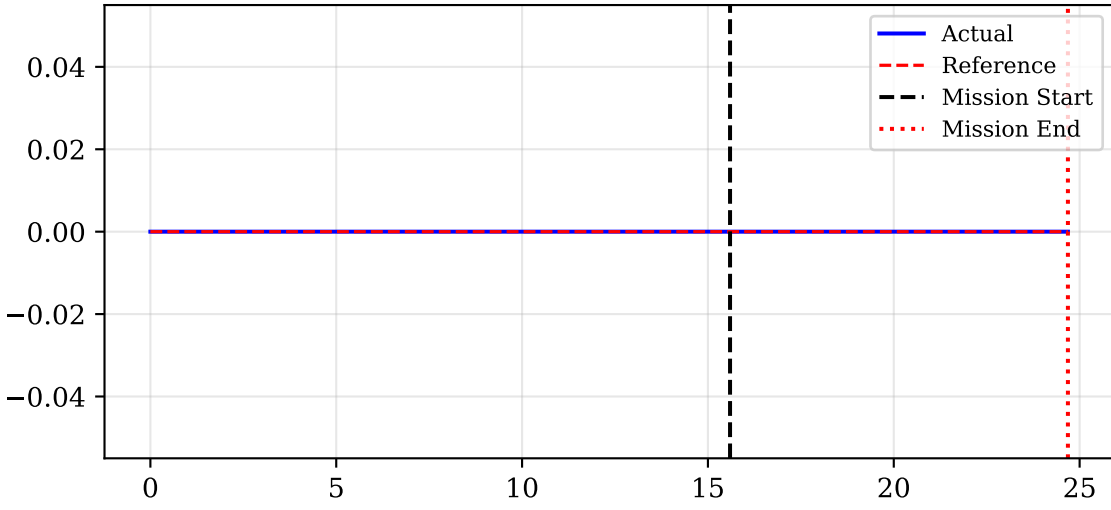
Peg Y [m]



Peg Z [m]

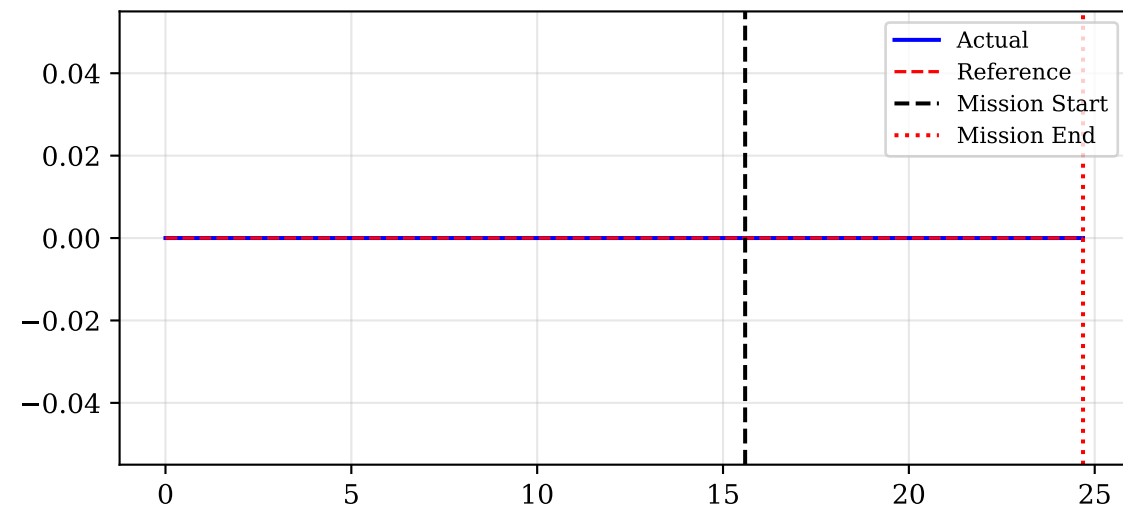


Peg Yaw [rad]

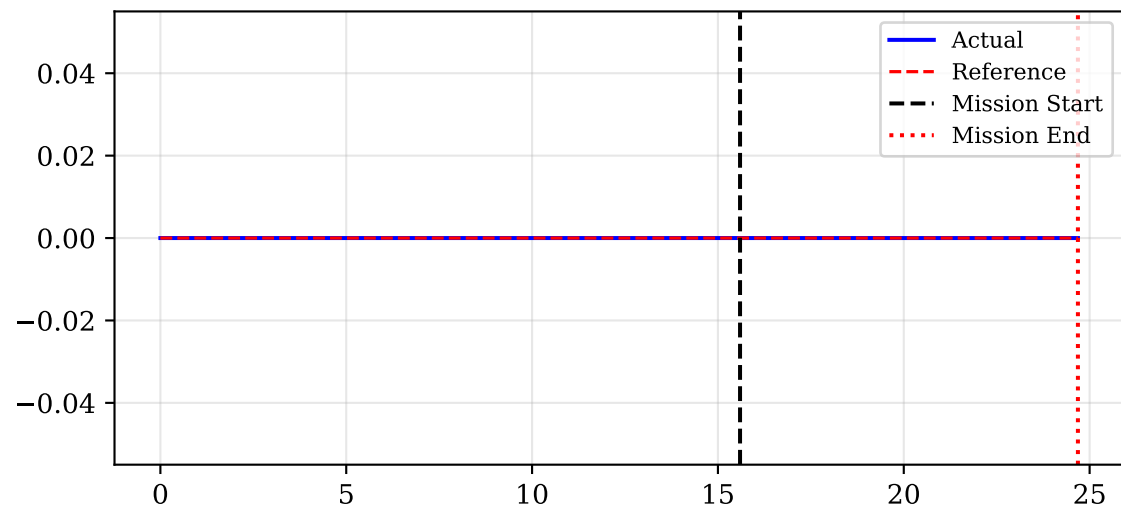


# Interaction Drone Velocities (ENU) and Yaw Rate

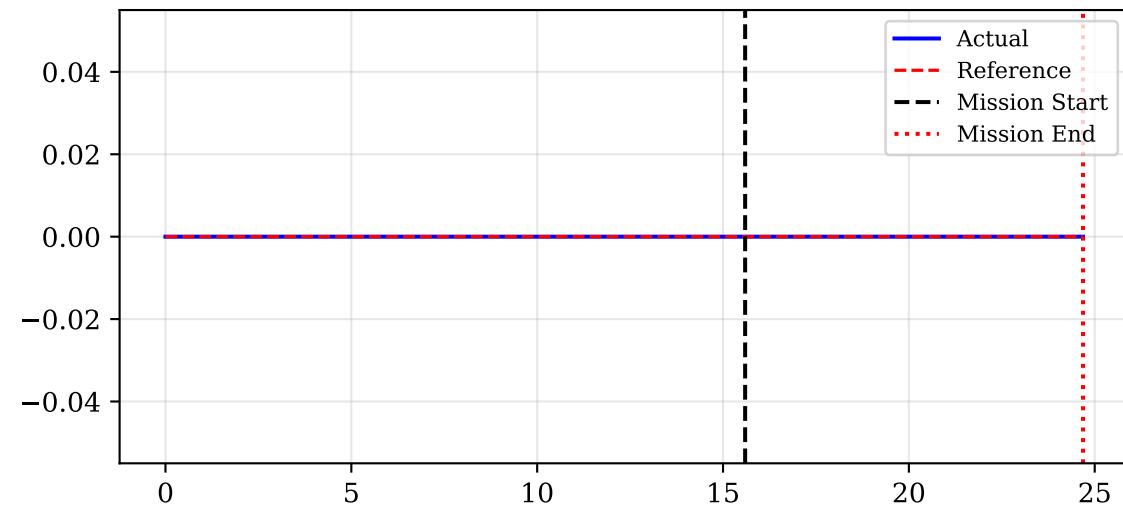
Vel X [m/s]



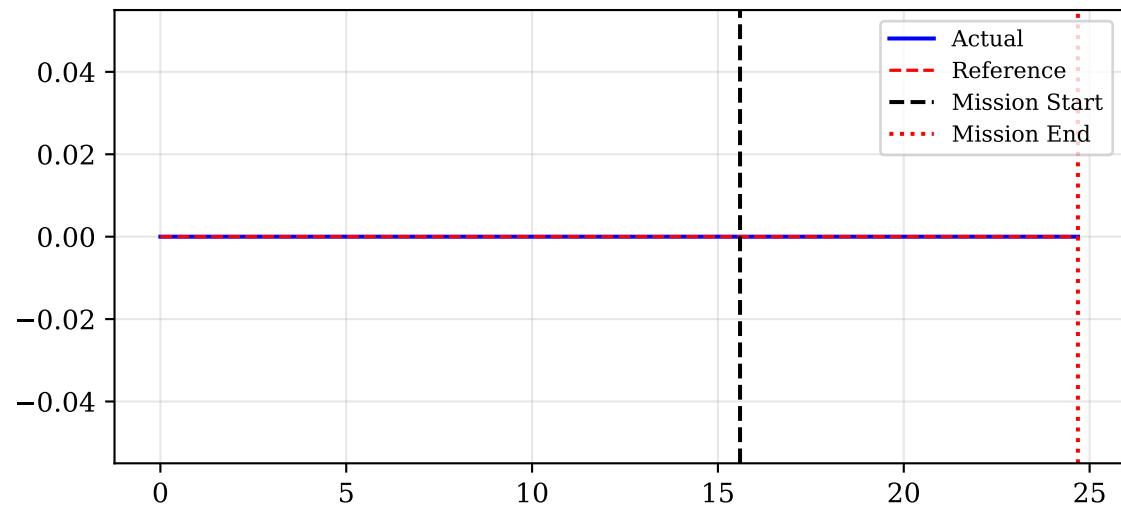
Vel Y [m/s]



Vel Z [m/s]

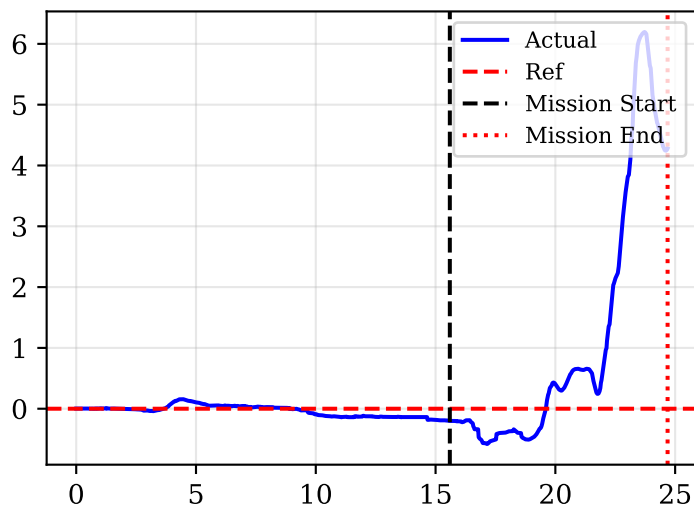


Yaw Rate [rad/s]

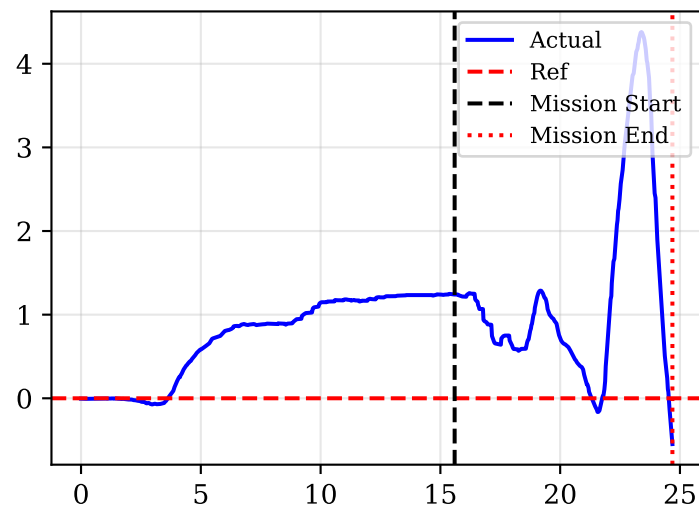


# Estimated Wrench (Momentum-Based Estimator)

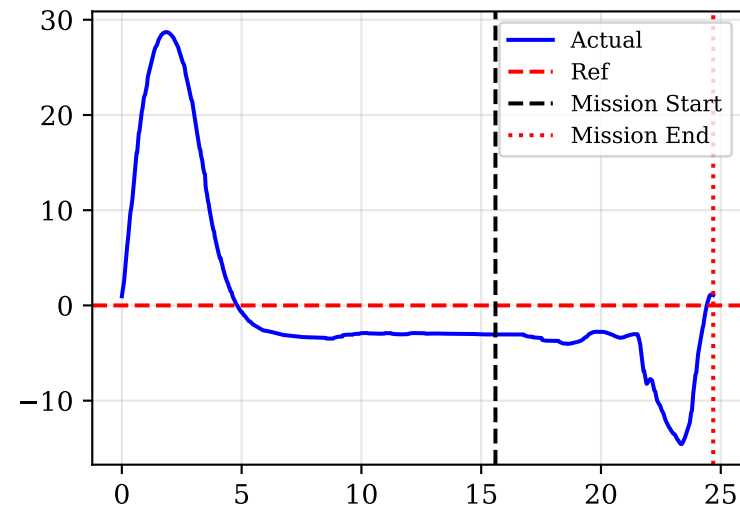
Force X [N]



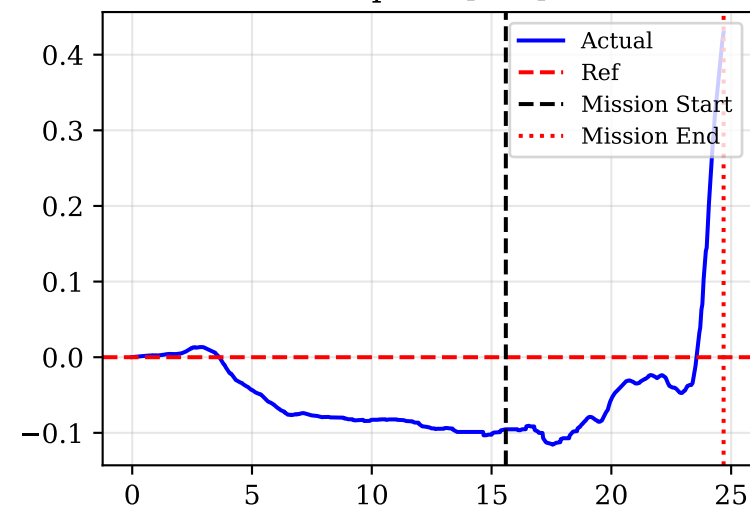
Force Y [N]



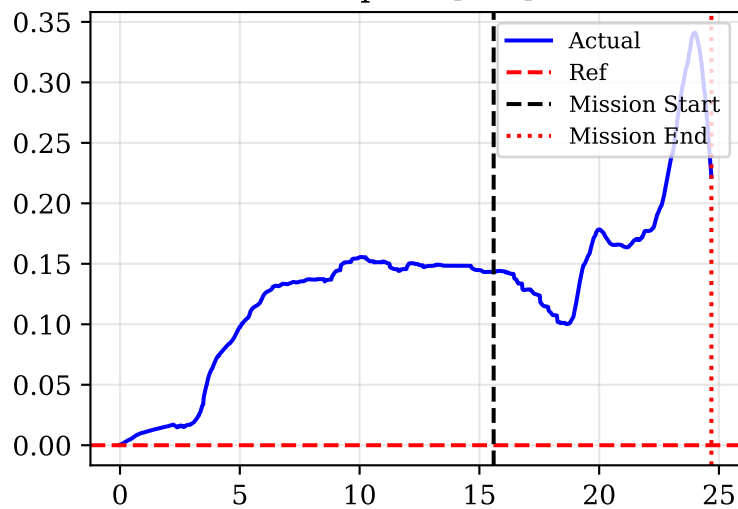
Force Z [N]



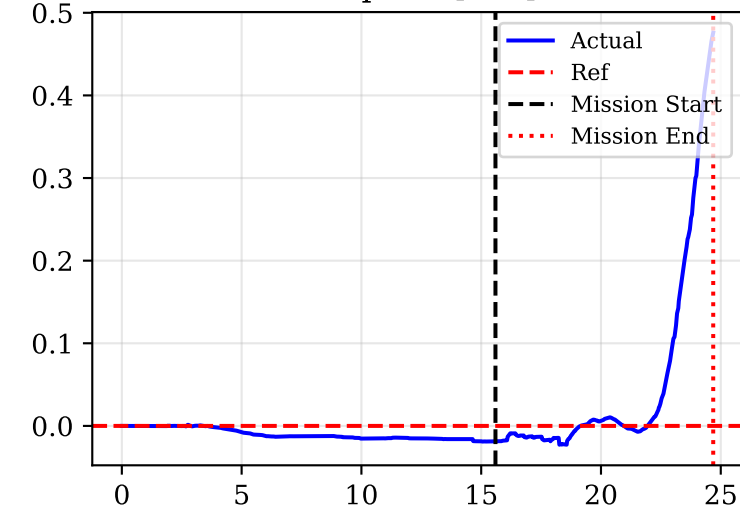
Torque X [Nm]



Torque Y [Nm]



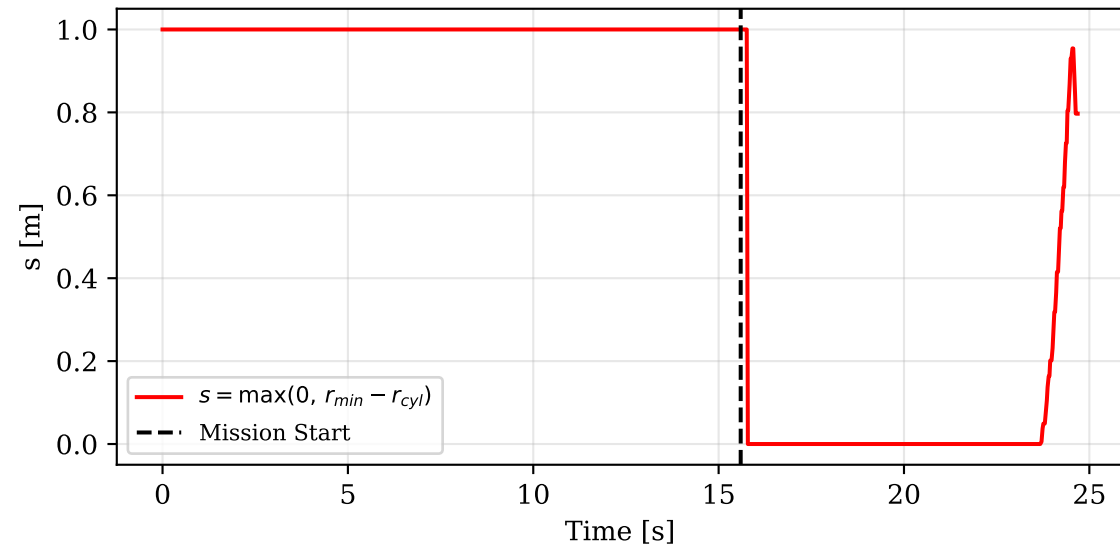
Torque Z [Nm]



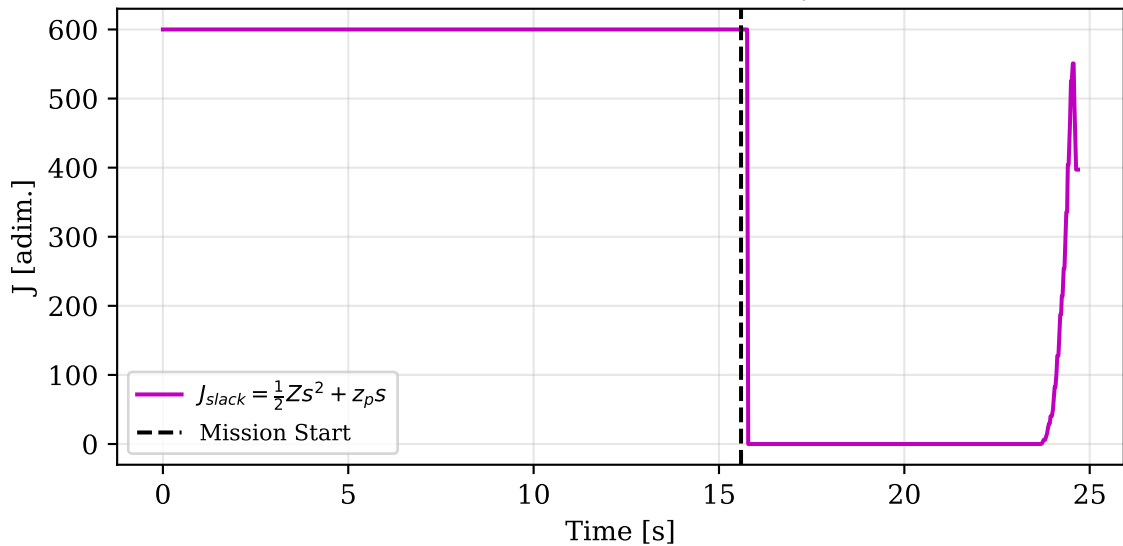


## Soft Constraint — Distanza Minima da Oggetto

Violazione geometrica  $s(t)$  [ $r_{min}=1.0$  m]

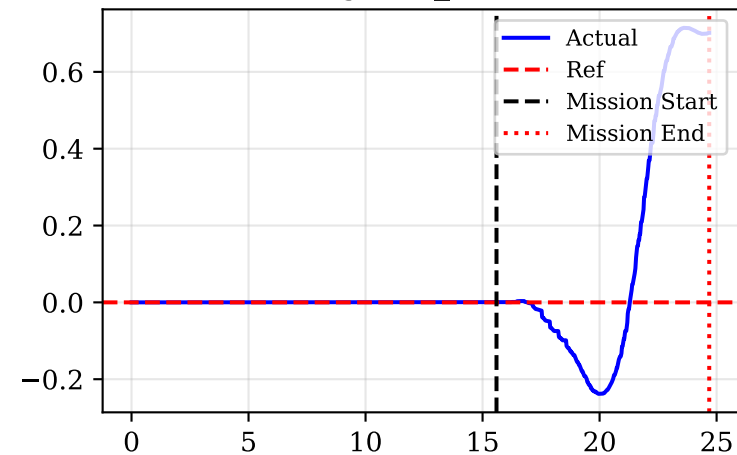


Costo slack  $J_{slack}(t)$  [ $Z=1e+03$ ,  $z_p=1e+02$ ]

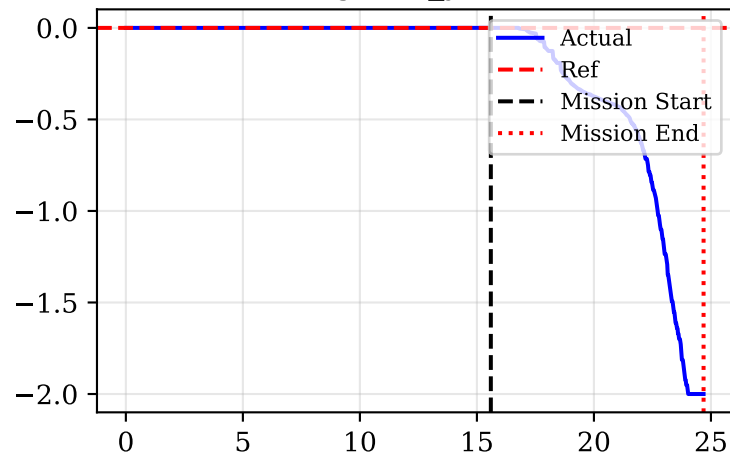


# Cartesian Integral Action (Anti-windup clipped)

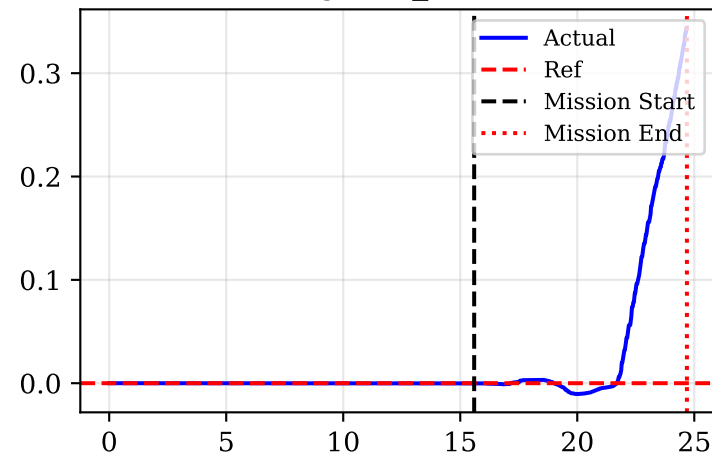
Integral  $e_x$  [m\*s]



Integral  $e_y$  [m\*s]

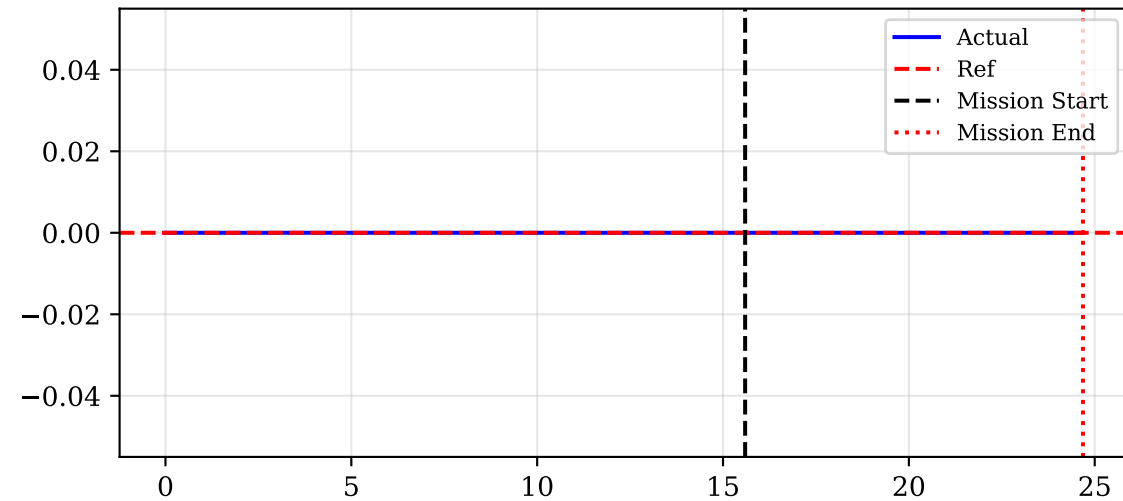


Integral  $e_z$  [m\*s]

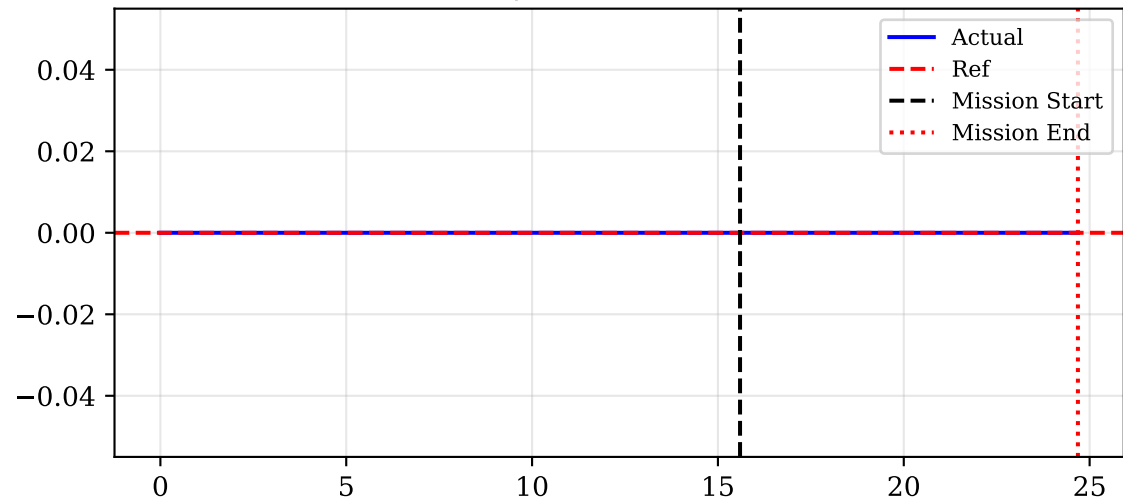


# Admittance Displacement in Sensor Frame

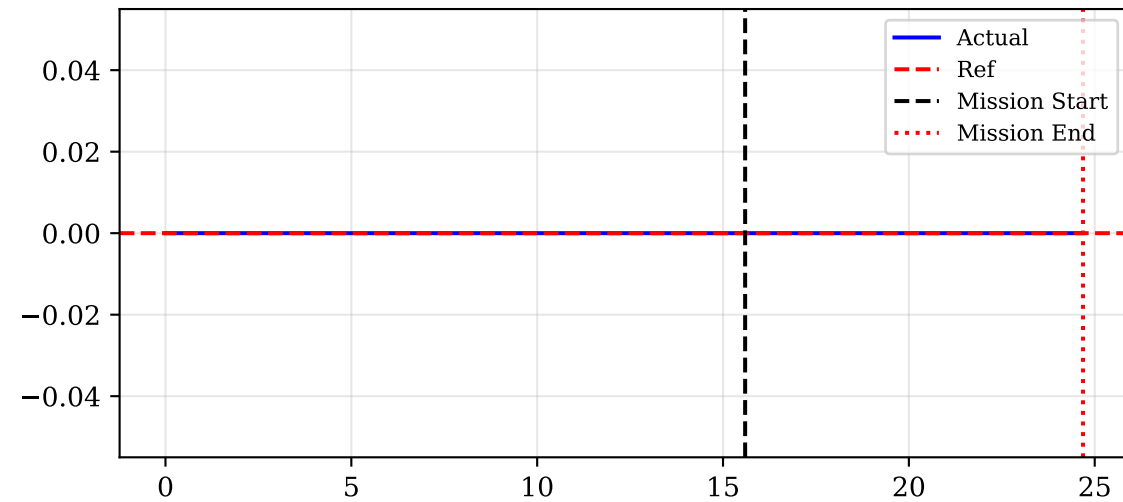
$\Delta p_{sx}$  [m] (Sensor X)



$\Delta p_{sy}$  [m] (Sensor Y)



$\Delta p_{sz}$  [m] (Sensor Z)



$\|\Delta p_s\|$  [m]

